



Cloud Computing

Assignment-Week 6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

A database stores customer records in encrypted form. However, an administrator accidentally changes several account balances due to a software bug. Which security objective has failed?

- A) Confidentiality only
- B) Integrity only
- C) Availability only
- D) Confidentiality and Availability

Correct Answer: B

Detailed Solution: Encryption protects confidentiality but does not prevent authorized users or software from incorrectly modifying data. The erroneous modification violates integrity.

QUESTION 2:

An employee logs into a fake VPN portal that perfectly imitates the company's official login page. Before any credentials are entered, which security property has already been compromised?

- A) Confidentiality
- B) Integrity
- C) Availability
- D) Authenticity

Correct Answer: D

Detailed Solution: The fake portal falsely claims to be the legitimate VPN service. The attack exploits **authenticity** before any confidential information is disclosed.

QUESTION 3:

Which of the following attacks is NOT considered a passive attack?



- A) Packet sniffing
- B) Traffic analysis
- C) Session hijacking
- D) Network monitoring

Correct Answer: C

Detailed Solution:

Session hijacking allows an attacker to take over an active communication session by injecting or modifying packets. Since it alters the communication, it is an active attack. Packet sniffing, traffic analysis, and network monitoring only observe communications without altering them and are therefore passive attacks.

QUESTION 4:

Which of the following attacks **cannot directly modify** the contents of transmitted data?

- A) Replay attack
- B) Man-in-the-Middle attack
- C) Eavesdropping
- D) Masquerading

Correct Answer: C

Detailed Solution:

Eavesdropping only captures data being transmitted and does not modify or inject messages. Hence, it is a passive attack.

QUESTION 5:

Which of the following best describes a Denial-of-Service (DoS) attack?

- A) An attacker secretly monitors network traffic without modifying it.
- B) An attacker floods a server with excessive requests, making the service unavailable to legitimate users.
- C) An attacker impersonates a legitimate user to gain unauthorized access.
- D) An attacker modifies data during transmission between two communicating parties.

Correct Answer: B



Detailed Solution: A Denial-of-Service (DoS) attack attempts to make a service or resource unavailable by overwhelming it with excessive traffic or resource requests. This is an active attack because it interferes with the normal operation of the system.

QUESTION 6:

Which of the following activities is least likely to be considered part of the reconnaissance phase?

- A) Performing DNS enumeration to identify subdomains
- B) Scanning the target to discover open ports and running services
- C) Exploiting a buffer overflow vulnerability to gain administrator privileges
- D) Searching public websites and social media for organizational information

Correct Answer: C

Detailed Solution:

Reconnaissance involves gathering information about the target using passive or active techniques. DNS enumeration, port scanning, and Open-Source Intelligence (OSINT) are common reconnaissance activities. Exploiting a buffer overflow is part of the exploitation (attack) phase, not reconnaissance.

QUESTION 7:

Statement I: Authentication determines whether a user is allowed to access a resource.

Statement II: Authorization determines what actions an authenticated user is permitted to perform.

Options:

- A) Statement I is TRUE and Statement II is FALSE.
- B) Statement I is FALSE and Statement II is TRUE.
- C) Both statements are TRUE.
- D) Both statements are FALSE.

Correct Answer: B

Detailed Solution: Authentication verifies who the user is, whereas authorization determines what that authenticated user is allowed to do.



QUESTION 8:

A financial institution must install a custom Linux kernel module, configure firewall rules at the operating system level, and deploy proprietary database software. However, it does not want to purchase or maintain physical servers. Which cloud service model is most appropriate?

- A) SaaS
- B) PaaS
- C) IaaS
- D) FaaS

Correct Answer: C

Detailed Solution: The organization requires control over the operating system and installed software, which is available in IaaS. The cloud provider manages the underlying physical infrastructure, while the customer manages the OS, middleware, and applications.

QUESTION 9:

An organization discovers that backups of its confidential data are automatically replicated to another country without its knowledge. Which cloud security concern has been violated?

- A) Data Location
- B) Recovery
- C) Virtualization
- D) Resource Pooling

Correct Answer: A

Detailed Solution: Automatic replication across geographic regions may violate legal or organizational requirements governing where sensitive data is stored.

QUESTION 10:

Which of the following cloud design practices is most effective in reducing vendor lock-in?

- A) Using provider-specific APIs whenever possible
- B) Developing applications using open standards and portable technologies



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- C) Storing data only in proprietary database formats
- D) Relying exclusively on managed proprietary services

Correct Answer: B

Detailed Solution: Open standards, containerization, and portable technologies reduce dependence on any single provider and simplify migration.